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Architecting Integrity: A Dual-Experience Blueprint for AI-Powered Proctoring Systems

Product Requirements Document (PRD)

The Aegis Interview Room is envisioned as the most secure and scientifically validated AI-powered interview environment, engineered to completely neutralize cheating vectors such as impersonation, ghostwriting, and unauthorized aid [1] [4]. Its core mission is to provide recruiters with an intuitive, actionable measure of candidate integrity while ensuring a frictionless and non-distracting experience for the candidate. This dual-experience principle is foundational to its design, mandating a separation of concerns where maximal information is provided to administrators and interviewers, while the candidate interface remains intentionally minimal to prevent cognitive overload and trigger false positives in biometric systems [69] [71]. The system's success is predicated on achieving a >98% detection rate for spoofing attacks, reducing false-positive flags to <2%, and maintaining 100% compliance with global biometric privacy laws like GDPR and BIPA [20].

The scope of the first version of the Aegis platform includes a web-based candidate interview portal, a real-time computer vision proctoring engine, a recruiter analytics dashboard, and secure storage for video artifacts [9] [80]. Key functional requirements are driven by the need for multi-modal behavioral biometrics. FR1 mandates Active Liveness Detection, where the system prompts candidates with randomized on-screen trajectories to validate pupillary response and micro-movements, effectively defeating 2D masks, 3D masks, and photo attacks [35] [36]. FR2 requires Continuous Passive Eye Tracking to monitor gaze vectors, fixation duration, and saccades, flagging prolonged off-screen gazes (>3 seconds) or erratic pupil dilation that may indicate reading from hidden scripts [2]. Advanced Gesture & Pose Recognition (FR3), utilizing skeletal tracking models like MediaPipe Pose, detects secondary persons in the frame, hand-to-ear gestures indicative of hidden earpieces, or unauthorized object manipulation [3] [38]. Further, Environmental & Device Scanning (FR4) uses object detection to identify secondary monitors or mobile phones, while Audio Forensics (FR5) analyzes background noise to detect secondary voices or keyboard typing cadence anomalies [4]. All this telemetry

feeds into an Integrity Scoring Engine (FR6), which aggregates the data into a single, understandable 0-100 score for recruiter review [8].

Non-functional requirements are stringent and directly tied to the system's integrity and user trust. Latency is a critical factor; real-time CV inference on the client-side must operate with less than 200ms latency to prevent stream desynchronization and maintain a seamless candidate experience [38] [100]. Accuracy targets are equally demanding, requiring facial recognition identity matches to exceed 99.5% and activity recognition models (DeepCNN) to achieve over 92% accuracy [48]. Privacy and security are paramount, given the processing of sensitive biometric data under regulations like GDPR and BIPA [18] [19]. All biometric data must be encrypted using AES-256 at rest and TLS 1.3 in transit, with PII masked in analytics dashboards. Data retention policies mandate automatic purges of raw data after 90 days to comply with data minimization principles [47]. Accessibility is also a core requirement, with the UI needing to be WCAG 2.1 AA compliant to serve all candidates, including those with motor or visual impairments [39] [40].

ID	User Story	Acceptance Criteria
US-01	As a candidate, I want to verify my camera and lighting before the interview starts so that the AI can calibrate properly.	System runs a 10-second calibration, checks lighting lux, and confirms facial landmark visibility before unlocking the "Start" button.
US-02	As a candidate, I want clear instructions if my face leaves the frame so I can correct it without being unfairly flagged.	A gentle, non-punitive UI overlay appears ("Please center your face") before a formal warning is logged.
US-03	As a recruiter, I want to see a timeline of "Risk Events" during the interview so I can quickly review suspicious moments.	Dashboard displays a scrub-able video timeline with red markers for high-risk events (e.g., "Off-screen gaze > 5s", "Secondary voice detected").
US-04	As a system admin, I want the system to automatically reject video-replay attacks.	System detects micro-inconsistencies in screen refresh rates (Moiré patterns) and analyzes audio-visual sync drift to block pre-recorded video feeds.

This structured approach ensures that every feature is designed to meet a specific objective, grounded in measurable criteria and user-centric scenarios, thereby building a robust framework for the system's development and evaluation.

Technical Requirements Document (TRD)

The technical architecture of the Aegis Interview Room is built upon a Hybrid Edge-Cloud model, a strategic choice that directly addresses the competing demands of low-latency processing and high-volume data analysis [38]. Heavy, latency-sensitive Computer Vision (CV) tasks run locally on the client's browser, leveraging WebAssembly (Wasm) and WebGPU for accelerated inference. This edge-first approach ensures that real-time telemetry streams remain responsive and synchronized, preventing the delays that could frustrate a candidate and degrade the quality of the biometric data being captured [100][114]. High-level aggregation, long-term storage, and the complex analytics presented to recruiters are handled by a scalable cloud backend. This division of labor optimizes both performance and privacy by minimizing the transmission of raw, sensitive video data to the cloud [101][145].

On the client side, the system utilizes WebRTC for high-fidelity, low-latency audio and video streaming, forming the foundation of the interview. The local inference engine, powered by frameworks like TensorFlow.js or ONNX Runtime Web, executes quantized ML models optimized for browser environments [9]. These include MediaPipe Face Mesh for 468 facial landmarks, a lightweight MPIIGaze CNN for gaze estimation, and MediaPipe Pose for 33-keypoint skeletal tracking [37][113]. A specialized liveness model tracks pupillary diameter changes in response to randomized screen brightness shifts to verify a live subject [133]. The server-side cloud architecture is built for scalability and resilience. An API Gateway using GraphQL provides flexible data fetching for the complex recruiter dashboard, while REST endpoints handle webhooks and other asynchronous communications. Real-time telemetry ingestion is managed by Apache Kafka, capable of handling the massive throughput from WebSocket streams containing biometric data at up to 60fps [9]. Processing Workers, built with Python and FastAPI, run computationally intensive algorithms for final report generation and anomaly detection [9].

Hardware and software requirements are specified for both client and server. Candidate clients require a minimum 720p webcam, a dual-core CPU equivalent to an i5/Ryzen 5, 8GB RAM, and a modern evergreen browser with WebGPU/WebGL support [38]. Network stability is crucial, with a minimum 5 Mbps upload speed and jitter below 100ms recommended [9]. The cloud infrastructure is provisioned on a Kubernetes cluster with auto-scaling capabilities. GPU nodes, such as NVIDIA A10G instances, are reserved for fallback heavy inference or post-interview forensic analysis [9]. Video artifacts are stored securely in AWS S3 buckets, which are configured for private access only, with all content

encrypted using AWS S3 Server-Side Encryption with Keys managed by AWS Key Management Service (SSE-KMS) [102][103].

Performance benchmarks are rigorously defined to ensure a reliable user experience. The CV models must process video at a minimum of 30 frames per second (FPS); if performance drops below 15 FPS, the system alerts the user to close background applications to improve performance [9]. To manage bandwidth, telemetry payloads streamed via WebSocket are compressed using MessagePack or Protobuf, keeping the average data transfer rate under 50 KB/s [9]. The cloud backend is designed for 99.99% uptime to ensure availability for all scheduled interviews. The system's security and compliance framework is multi-layered. It adheres to Illinois' Biometric Information Privacy Act (BIPA) and Texas' CUBI, which require written consent and specific data handling policies [24]. Biometric templates, such as facial geometry, are hashed and never stored in plaintext, preventing reverse-engineering [144]. Anti-spoofing measures are defense-in-depth, combining Moiré pattern detection and audio-visual sync analysis to block video replay, and remote photoplethysmography (rPPG) to detect a real heartbeat in the video feed, countering deepfake attacks [35][133].

Application Flow and Monitoring Logic

The Aegis Interview Room orchestrates a three-phase process: Pre-Interview Onboarding & Calibration, Interview Execution & Continuous Monitoring, and Post-Interview Reporting. This flow is meticulously designed to establish a baseline of normal behavior, vigilantly detect deviations, and present findings in an actionable format. The entire journey is guided by the asymmetric information principle, where the candidate's path is streamlined and distraction-free, while the system performs exhaustive background analysis.

In Phase 1, Pre-Interview Onboarding & Calibration, the system ensures the candidate's environment and identity are verified before the interview begins. This starts with authentication via a secure magic link, followed by a mandatory Hardware Check that verifies the functionality of the webcam, microphone, and internet connection [9]. Next, an Environment Scan is conducted where the candidate is prompted to pan their camera around the room; an object detection model like YOLOv8 scans for secondary monitors, mobile phones, or unauthorized individuals [3]. The cornerstone of this phase is the Active Liveness Challenge. A moving dot appears on a randomized trajectory, and the

candidate is instructed to follow it with their eyes alone [133]. The system validates this by analyzing pupillary response and 3D depth mapping, confirming a live human presence and defeating static photos or masks [35]. Finally, a Baseline Calibration period records the candidate's resting gaze, skin tone, and baseline lighting to normalize future anomaly detection, ensuring that subsequent checks are context-aware [9].

Phase 2, Interview Execution & Continuous Monitoring, is the core operational period where the AI proctoring engine is fully engaged. Question delivery is managed by either the AI or a human interviewer [4]. Simultaneously, passive telemetry streams continuously from the candidate's browser. The Eye Tracking module monitors fixation points; if the candidate's gaze shifts to a location like the top-right corner—a common spot for hidden notes—for more than three seconds, a low-level flag is raised [2]. The Head Pose estimator tracks yaw, pitch, and roll; a turn exceeding 45 degrees triggers an immediate audio mute and a warning on-screen to prevent the candidate from consulting external materials [66]. The Gesture Tracking system uses skeletal models to monitor hand movements, raising a high-level flag if hands leave the frame or approach the ear, indicating a potential hidden communication device [3]. Every 5-10 minutes, the system initiates Randomized Active Checks, such as flashing a color or asking for a micro-action (e.g., "blink twice"), to re-verify liveness and thwart sophisticated video injection or deepfake attacks [133]. Throughout the session, Audio Forensics analyze background sounds, detecting keyboard typing that doesn't match speech cadence or whispers from a secondary person [4].

Phase 3, Post-Interview Reporting, transforms raw telemetry into actionable intelligence for the recruiter. Upon session termination, the client-side CV engine finalizes the local telemetry log and uploads a compressed summary to the cloud [9]. Cloud processing workers then aggregate this data, applying the "Integrity Scoring Algorithm" to generate a holistic assessment [8]. The final output is a richly annotated recruiter dashboard. The video is indexed with telemetry overlays, and timestamped Risk Events are plotted on a scrub-able timeline [9]. The recruiter can click on any red flag marker, which instantly jumps the video to that moment and pauses it, allowing them to investigate the context. A side-panel displays the exact CV data that triggered the alert, providing transparency and evidence for their judgment [8].

UI/UX Design and Asymmetric Experience Architecture

The UI/UX design of the Aegis Interview Room is fundamentally shaped by the asymmetric information architecture, creating two distinct but complementary user experiences: a minimalist, distraction-free "Interview Room" for the candidate and a data-rich, analytical "Command Center" for the administrator and interviewer. This separation is not merely aesthetic but a core design principle essential for the system's effectiveness and ethical operation.

The Candidate Interface, or "Interview Room," is governed by the principle of minimal cognitive load. Complex UI elements can induce erratic eye movements, which would be incorrectly flagged as suspicious by the gaze-tracking engine, leading to false positives [69] [71]. Therefore, the design is exceptionally clean and sparse. The default dark mode reduces eye strain and enhances contrast for the pupillary tracking algorithms [99]. The primary screen layout centers the interviewer's video feed or question text, with the candidate's own small Picture-in-Picture webcam feed in the bottom right [9]. A subtle top bar displays only essential status indicators: a timer, internet connection strength, and a discreet "Privacy/Recording" badge. During the initial calibration, a faint, semi-transparent oval guide helps the candidate position themselves correctly, but this element disappears entirely once the interview begins to avoid distracting the user [9]. The system's feedback mechanism is deliberately non-punitive. Instead of a jarring red alert for minor infractions, a soft, glowing amber border appears around the screen with a gentle prompt like, "Please keep your face within the frame," giving the candidate a chance to self-correct before a formal risk event is logged [9]. When an active liveness check is needed, a smooth, non-intrusive modal slides in, explaining the simple action required [9].

In stark contrast, the Recruiter/Admin Interface, the "Command Center," is designed for rapid, data-driven decision-making. Its layout is divided into panels: a large left panel for the main video player and a compact right panel for analytics [80]. The centerpiece of the right panel is a large circular gauge displaying the candidate's final Integrity Score (e.g., 94/100 - "High Integrity"), providing an immediate, at-a-glance assessment of the session's integrity [9]. Below the score is a multi-layered telemetry timeline that allows the recruiter to visualize different data streams simultaneously [8]. Layer 1 might show a gaze heatmap revealing where the candidate looked most frequently, Layer 2 could display a graph of head pose angles to identify significant turns away from the screen, and Layer 3 might highlight spikes in background audio, such as keyboard typing [9]. A powerful feature is the "Ghost View" toggle, which superimposes the real-time CV data—

such as the MediaPipe skeletal wireframe and gaze vectors—directly onto the video player, allowing the recruiter to see exactly what the AI saw at any given moment [33]. This visualization bridges the gap between raw data and human understanding. Clicking a red flag on the timeline instantly navigates the video to that timestamp and pauses it, enabling efficient investigation of specific risk events [9].

Accessibility and inclusivity are woven into both interfaces. The candidate flow must be fully navigable via keyboard (`Tab` and `Enter`) and compatible with screen readers, using appropriate ARIA labels for all dynamic proctoring status updates [39] [67]. Crucially, the system includes an "Accommodation Mode" [29]. If a candidate discloses a medical condition like Tourette's syndrome or a visual impairment that makes strict proctoring impractical, the recruiter can activate this mode. This disables sensitive tracking like gaze and micro-expression analysis, relying instead on less intrusive methods like identity verification and environmental scanning, thus balancing security needs with accessibility requirements [29] [42]. Both the candidate and recruiter interfaces must adhere to WCAG 2.1 AA standards, ensuring they are usable by people with a wide range of abilities [44]. This includes using sufficient color contrast, providing text alternatives for non-text content, and ensuring interactive elements are easy to activate, complying with guidelines like Success Criterion 2.5.8 (Target Size) [135][139].

Backend Data Architecture and Security Schema

The Aegis Interview Room employs a polyglot persistence strategy, recognizing that its diverse data types—from structured metadata to high-frequency biometric telemetry—require specialized storage solutions for optimal performance, scalability, and query efficiency [7]. This approach involves four distinct database technologies working in concert. Relational data, such as user profiles, session metadata, permissions, and billing information, is managed by PostgreSQL, chosen for its robust transactional integrity and ACID compliance [55] [84]. High-velocity, time-stamped biometric telemetry—capturing data like gaze vectors and head pose at 30-60fps—is stored in a dedicated Time-Series database, specifically TimescaleDB, which extends PostgreSQL to provide automatic partitioning and optimized time-based queries [7]. Large binary objects, including full-resolution video recordings, audio files, and generated reports, are stored in a secure, encrypted object store, such as AWS S3 [102]. Finally, for identity verification and deepfake detection, the system uses a Vector Database like Pinecone or Milvus to store facial

embedding vectors, enabling fast and accurate similarity searches to confirm a candidate's identity against a known template [48].

The relational schema for PostgreSQL forms the backbone of the system's metadata. The `users` table stores candidate, recruiter, and admin profiles, linking each user to a cryptographic hash of their facial embedding rather than the raw vector, a critical security measure to prevent reverse-engineering [21]. The `sessions` table tracks all interview activities, connecting a candidate and a recruiter, recording the job role, session status, and the final aggregated `integrity_score` [15]. The `risk_events` table is the audit log of the proctoring system, storing every flagged incident with a precise timestamp, event type (e.g., 'gaze_offscreen'), severity level, confidence score, and a JSONB field for detailed, flexible metadata about the specific event [9].

The schema for the TimescaleDB hypertable, `biometric_telemetry`, is optimized for write throughput. It is partitioned by time, allowing for efficient storage and querying of millions of data points per session. This structure is essential for running historical analyses and training machine learning models on long-term behavioral patterns [7]. The following SQL code illustrates the creation of this hypertable:

```
-- Telemetry Data (Hypertable)
CREATE TABLE biometric_telemetry (
    time TIMESTAMPTZ NOT NULL,
    session_id UUID NOT NULL,
    gaze_pitch DECIMAL,
    gaze_yaw DECIMAL,
    head_roll DECIMAL,
    head_pitch DECIMAL,
    head_yaw DECIMAL,
    pupil_diameter_mm DECIMAL,
    blink_status BOOLEAN
);

-- Convert to TimescaleDB hypertable for automatic partitioning by time
SELECT create_hypertable('biometric_telemetry', 'time');
```

Data security and privacy are embedded into the architecture's core. All Personally Identifiable Information (PII) is encrypted at the application layer before being written to the database. Biometric data is protected through multiple layers of defense. Raw facial embeddings are stored exclusively in the vector database, while the relational database only holds a one-way hash, rendering the original data irrecoverable even if the database

is compromised [144]. Video files in AWS S3 are encrypted using SSE-KMS, meaning encryption keys are managed by a dedicated key management service, not the application [102]. Access to these private S3 buckets is strictly controlled, granted only via pre-signed URLs that expire after a short duration (e.g., 15 minutes), ensuring that recorded sessions cannot be accessed without explicit authorization [9]. The system enforces a strict data retention policy. Automated processes, likely implemented as AWS Lambda functions or cron jobs, are scheduled to permanently delete raw `biometric_telemetry` tables and video files from S3 after 90 days, retaining only the anonymized summary data in the relational database for long-term analytics and compliance purposes, aligning with GDPR's storage limitation principle [47].

Implementation Plan and Risk Management

The implementation of the Aegis Interview Room follows a six-month roadmap to General Availability (GA), structured into five distinct phases that systematically build and validate the platform's components. Each phase has clear milestones and deliverables, supported by a dedicated cross-functional team.

Phase	Duration	Key Milestones & Deliverables
Phase 1: Discovery & Core ML Prototyping	Weeks 1-6	Milestone 1.1: Finalize PRD/TRD and UI/UX wireframes. Milestone 1.2: Train and optimize lightweight CV models for WebAssembly/ WebGPU deployment. Milestone 1.3: Develop the Active Liveness algorithm. **Deliverables:** Working local prototype of the CV engine running in a browser at 30+ FPS.
Phase 2: Backend & Infrastructure Setup	Weeks 7-12	Milestone 2.1: Provision cloud infrastructure (Kubernetes, TimescaleDB, PostgreSQL, S3). Milestone 2.2: Develop WebRTC signaling servers. Milestone 2.3: Build the WebSocket telemetry ingestion pipeline (Kafka -> TimescaleDB). Milestone 2.4: Implement security protocols (Encryption, BIPA compliance). **Deliverables:** Fully functional, secure backend APIs and database schemas.
Phase 3: Full-Stack Integration & UI Development	Weeks 13-18	Milestone 3.1: Build Candidate Interview Room UI and integrate local CV engine. Milestone 3.2: Build Recruiter Command Center dashboard. Milestone 3.3: Integrate the Integrity Scoring Algorithm. Milestone 3.4: Implement "Ghost View" and timeline overlay. **Deliverables:** End-to-end working application (Alpha version).
Phase 4: Adversarial Testing & Beta	Weeks 19-22	Milestone 4.1: Red Team engagement to attempt bypasses (video replay, 3D masks, deepfakes). Milestone 4.2: Bias & Fairness Audit across diverse demographics. Milestone 4.3: Closed Beta with 50 clients and 500 candidates for UX feedback. **Deliverables:** Hardened, bias-audited, beta-tested platform.
Phase 5: Launch & Scaling	Weeks 23-24	Milestone 5.1: Production deployment with auto-scaling enabled. Milestone 5.2: Finalize documentation, API keys, and SDKs for enterprise integration. Milestone 5.3: General Availability (GA) Launch. **Deliverables:** Publicly available, production-ready platform.

Resource allocation is critical to this timeline, with a lean but highly specialized team. The plan includes a Team Lead / Architect, three Computer Vision / ML Engineers, two Backend Engineers, two Frontend Engineers, one UX/UI Designer, one part-time Security / Compliance Specialist, and two QA / Red Team engineers [8] [9]. This structure ensures that expertise in each critical domain is represented throughout the development lifecycle.

A proactive risk management strategy is embedded in the plan, addressing the most significant threats to the project's success. The primary risk is a high false positive rate, which could damage candidate trust and render the system unusable [69]. The mitigation strategy involves implementing a contextual AI with a "grace period" for minor infractions and requiring human review for any session with an integrity score below a predefined threshold (e.g., 80) [110]. The second major risk is client hardware limitations, as the system's performance depends on the candidate's device [5]. The contingency plan is a "fallback mode": if WebGPU fails or FPS drops below a critical threshold, the system

gracefully degrades, relaxing proctoring rules (e.g., disabling gaze tracking) to maintain session continuity [38]. Third, there is a critical risk of privacy backlash and legal challenges due to the processing of biometric data under regulations like GDPR and BIPA [18] [19]. Mitigation involves prioritizing privacy by design, performing all sensitive processing on-device whenever possible to minimize data transmission, and engaging legal counsel early to ensure all consent mechanisms and data handling practices are compliant [24] [101]. Finally, the threat landscape for deepfake and AI spoofing is constantly evolving, representing a critical, ongoing arms race [133]. The plan allocates 15% of post-launch sprint capacity to continuously update and retrain the anti-spoofing models, ensuring the system's defenses remain effective over time [9].

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